

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 90-271

WASTE DISCHARGE REQUIREMENTS
FOR
CITY OF CHOWCHILLA
WASTEWATER TREATMENT FACILITY
MADERA COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The City of Chowchilla (hereafter Discharger) owns and operates a wastewater treatment and disposal facility in Madera County. The wastewater treatment and disposal facility is in Section 31, T9S, R16E, MDB&M, as shown in Attachment A, which is a part of this Order.
2. Waste Discharge Requirements Order No. 72-178, adopted by the Board on 25 February 1972, prescribes requirements for a discharge from the treatment facility to evaporation/percolation ponds and to adjacent lands for surface irrigation of fodder, fiber and seed crops not intended for human consumption.
3. Wastewater Reclamation Requirements Order No. 87-035, issued to Bill Jennings, regulates use of reclaimed water from the treatment facility on 25 acres of adjacent agricultural land.
4. Order No. 72-178 is neither adequate nor consistent with current plans and policies of the Board.
5. The Discharger discharges approximately 800,000 gallons per day from a secondary treatment facility to evaporation/percolation ponds. Wasted sludge is dried and hauled to landfill sites for final disposal.
6. The facility consists of two inlets, one for municipal and the other for industrial discharge. Currently, the treatment facility is not receiving industrial waste. The treatment plant is split into two treatment trains, each consisting of a grit chamber, a primary floatation sedimentation basin, an activated sludge unit, and a secondary clarifier. The effluent is then sent to a final clarifier before being discharged to six evaporation/percolation ponds. Sludge is processed by means of two sludge digesters and eight sludge drying basins.
7. Each treatment train has a design capacity of 0.9 million gallons per day for a total plant capacity of 1.8 million gallons per day. Disposal capacity of the ponds has not been verified.
8. Depth to first groundwater is approximately 80 feet with a gradient to the southeast. The site lies within the Madera Hydrologic Area (No. 545.20) in the San Joaquin Valley Floor Hydrologic Unit, as defined on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.

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9. The Board, on 25 July 1975, adopted a Water Quality Control Plan for the San Joaquin Basin (5C), hereafter "Basin Plan". These requirements are consistent with the plan.
10. Local water quality is good (EC = 580 umhos/cm). The beneficial uses of underlying ground water domestic, industrial, and agricultural supply.
11. Surface water drainage is to Berenda Slough, a tributary of the San Joaquin River.
12. The beneficial uses of the San Joaquin River are municipal, industrial, and agricultural supply; recreation; esthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydroelectric power generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.
13. The action to update waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
14. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
15. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.
16. The Board adopted Order No. 90-180 on 22 June 1990 regulating the discharge. The Order was inadvertently adopted without revisions the staff intended to include.

IT IS HEREBY ORDERED that Order No. 90-180 is rescinded and City of Chowchilla, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at the above-described facility:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.

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3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Title 23, CCR, Section 2521(a) and 2522(a), is prohibited.

B. Discharge Specifications:

1. The monthly average daily discharge flow shall not exceed 1.8 million gallons.
2. The effluent from the treatment facility shall not contain constituents in excess of the following:

<u>Constituent</u>	<u>Unit</u>	<u>Mean</u>	<u>Maximum</u>
BOD ₅ ¹	mg/l	40	80
Total Suspended Solids	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5

¹ Five-day, 20° Celsius biochemical oxygen demand

3. The maximum specific electrical conductivity (EC) of the discharge shall not exceed 900 umhos/cm.
4. The evaporation/percolation ponds shall not have a pH less than 6.5 or greater than 9.0.
5. The dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
6. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
7. Ponds shall be managed to prevent breeding of mosquitos. In particular,
 - a. An erosion control program shall assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

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8. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
9. Use of reclaimed water shall be limited to surface irrigation of orchards and vineyards, and fodder, fiber, and seed crops.
10. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Uses for reclaimed water other than those identified herein shall be subject to the approval of the Executive Officer and shall comply with Title 22, CCR, Division 4.
11. The treatment facilities shall designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
12. The evaporation/percolation ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than 1 foot (measured vertically).

C. Sludge Disposal

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Title 23, CCR, Section 2510, et seq. (Chapter 15), and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 14 days in advance of the change.

D. Receiving Water Limitations

The proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 14 days in advance of the change.

1. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified. (For purposes of comparison, background water quality shall be determined when background monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards.")

2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in Title 22, CCR, Division 4, Chapter 15.
3. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven day period.
4. Exceed concentrations of radionuclides specified in Title 22, CCR, Division 4, Chapter 15.
5. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
6. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

E. Provisions

1. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this Order.
2. The Discharger shall submit technical reports as directed by the Executive Officer.
3. The Discharger shall comply with the attached Monitoring and Reporting Program No. 90-271, and any revisions thereto as ordered by the Executive Order.
4. The Discharger shall comply with the "Standards Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 July 1990, which is a part of this Order.
5. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
6. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
7. In addition to modification of the Order under circumstances described in Standard Provision A.3, this Order may be modified if sludge use or disposal practices change, if applicable management practices or numerical limitations for pollutions in sewage sludge

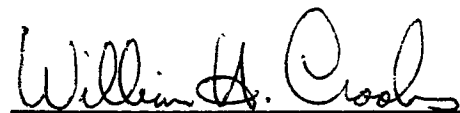
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necessitate it, if there is endangerment to human health or the environment, or if the Board elects to review and revise the Order on its own motion.

8. The Discharger shall provide a letter report within 90 days of the effective date of this Order, identifying the names, titles, and certificate grades of persons supervising and operating the facility, as required by Standard Provision E.1, and identifying who to contact in an emergency.
9. The Discharge shall submit the technical reports described in Standard Provisions A.9 and B.2 and a copy of the operation and maintenance manual for the facility within 90 days of the effective date of this Order.
10. The Discharge shall submit the following reports within 90 days of the effective date of this Order:
 - a. A report verifying the ponds are designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency pursuant to Specification B.11.
 - b. A report verifying that the ponds will have sufficient freeboard to accommodate allowable wastewater flow and design seasonal precipitation pursuant to Specification B.12.
11. The Discharger must comply with all conditions of the Order. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in modification or revocation of this Order.
12. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 28 September 1990.


WILLIAM H. CROOKS, Executive Officer

JGW:fmc:9/28/90

Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 90-271
FOR
CITY OF CHOWCHILLA WASTEWATER TREATMENT FACILITY
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Specific sample locations shall be established under direction of the Board's staff and a description of the stations shall be attached to this Order.

INFLUENT MONITORING

Samples shall be collected at approximately the same time as effluent samples and should be representative of the influent for the period sampled. The following shall constitute the influent monitoring program:

<u>Constituent</u>	<u>Unit</u>	<u>Measurement</u>	<u>Frequency</u>
Flow	MGD	Meter	Daily
pH	pH units	Grab	Daily
Specific Electrical Conductivity @25°C	umhos/cm	Grab	Daily
Total Suspended Solids	mg/l	24-hr. Composite	Monthly
BOD ₅ ¹	mg/l	24-hr. Composite	Monthly

¹ Five-day, 20° Celsius biochemical oxygen demand.

EFFLUENT MONITORING

Effluent samples (except for Dissolved Oxygen) shall be collected just prior to discharge to the evaporation/percolation ponds. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from disposal ponds will be considered adequately composited. The following shall constitute the effluent monitoring program:

<u>Constituents</u>	<u>Unit</u>	<u>Measurement</u>	<u>Frequency</u>
Flow	MGD	Meter	Daily
Settleable Matter	ml/l	Grab	Daily
Specific Electrical Conductivity @ 25°C	umhos/cm	Grab	Daily

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<u>Constituents</u>	<u>Unit</u>	<u>Measurement</u>	<u>Frequency</u>
Suspended Matter	mg/l	24-hr. Composite	Weekly
BOD ₅	mg/l	24-hr. Composite	Weekly
Dissolved Oxygen ²	mg/l	Grab	Weekly
pH	pH units	Grab	Weekly
Nitrate (N)	mg/l	Grab	Monthly
Standard Minerals ³	mg/l	Grab	2/year

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- ¹ If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly mean for BOD), or indicate a violation and potential upset of the treatment process, the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.
- ² Samples shall be collected at a depth of one foot from each pond, opposite of the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.
- ³ Standard mineral analyses shall include calcium, carbonate, chloride, fluoride, iron, magnesium, potassium, sodium, sulfate, total dissolved solids, and total phosphorous.

POND MONITORING

The freeboard shall be monitored on all evaporation/percolation ponds in use to the nearest half-foot. The following shall constitute the freeboard monitoring program:

<u>Constituent</u>	<u>Unit</u>	<u>Type of sample</u>	<u>Frequency</u>
Freeboard	feet	observation	Weekly

Permanent markers shall be placed in each pond with calibrations indicating the water level at design capacity and available operational freeboard.

In addition, the Discharge shall inspect the condition of the pond once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharge shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

SOURCE WATER MONITORING

A representative sample shall be collected from the source which supplies water to the facility according to the specified frequency.

<u>Constituent</u> ¹	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Standard Minerals	mg/l	Grab	Annually

¹ If the source water is from more than one source, the EC shall be reported as the weighted average and include copies of supporting calculations.

SLUDGE MONITORING

Annually, a composite sample of wasted sludge shall be analyzed, on a dry weight basis, for Total Solids (%), Nitrogen (Total, NH₄-N, and NO₃-N), Total Phosphorous, Total Potassium, Total PCBs, and totals of specific metals (Pb, Zn, Cu, NI, CD, and Ag). Analytical results shall be submitted to the Executive Officer. Analysis of soluble concentrations of these specific metals shall also be included. If final disposal is proposed to go to land, a technical report analyzing application rates and procedures relative to Department of Health Services' Manual of Good Practices for Landspreading of Sewage Sludge and EPA's Process Design Manual for Land Application of Municipal Sludges and Title 23, California Code of Regulations, Section 2511(f), shall be completed and submitted to the Executive Officer of approval.

GROUND WATER MONITORING

By 1 October 1990, the Discharger shall develop a ground water monitoring network consisting of one or more background monitoring wells and three or more downgradient wells. All well locations and construction features are subject to the prior approval of the Executive Officer and must be sufficient to monitor potential impacts of the disposal operation on the uppermost ground-water aquifer.

Samples shall be taken **quarterly** from approved background monitoring well(s) for one year and analyzed for the parameters specified below. Data from these analyses shall be reported to the Board by 1 October 1991 for use in determining water quality protection standards.

If subsequent sampling of background monitoring wells indicate significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste disposal activities, the Discharge may request modification of the water quality protection standards.

The downgradient wells shall constitute "points of compliance (POC)". In conjunction with background monitoring, monitoring of POC's will enable one to determine compliance with water quality protection standards. The ground water surface elevation (in feet and hundredths, M.S.L.) in wells shall be measured on a quarterly basis and used to determine the gradient and direction of ground water flow. This information shall be displayed on a water flow net diagram for the site. Water samples shall be collected from wells in the approved monitoring network and analyzed as follows:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Standard Minerals ¹	mg/l	Grab	Annually

¹ Standard mineral analyses shall include calcium, carbonate, chloride, fluoride, iron, magnesium, nitrate, potassium, sodium, sulfate, total dissolved solids, specific electrical conductance, pH, and total phosphorous.

Following each sampling event, the discharger shall determine whether there is statistically significant increase over water quality protection standards for each parameter and consistent analyzed. If the discharger or the Board finds there is a statistically Significant increase in indicator parameters or waste constituents over the water quality protection standards at the POC's, the Discharger shall notify the Board, or acknowledge the Boards findings, and submit, within 90 days, either a technical report within 90 days. The report

must contain a plan and time schedule for implementing a corrective action program designed to achieve compliance with water quality protection standards.

REPORTING

Monthly monitoring reports shall be submitted to the Regional Board by the 15th day of the following month.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By **31 January** of each year, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names and general responsibilities of persons operating and maintaining the wastewater treatment plant.
- b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4).
- d. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when the documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

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All reports submitted in response to this Order shall comply with the signatory requirements of Standard Provision B.3.

The Discharger shall implement this monitoring program beginning the first of the month following the date of signature below.

Ordered by

William H. Crooks
WILLIAM H. CROOKS, Executive Officer

28 September 1990
(Date)

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CITY OF CHOWCHILLA WUTF
MADERA COUNTY

INFORMATION SHEET

The City of Chowchilla owns and operates a wastewater treatment and disposal facility in Section 31, T9S, R16E, MDB&M. Average wastewater flow is approximately 0.8 mgd.

The Board, on 25 February 1972, adopted Order No. 72-178, which prescribes requirements for a discharge from the treatment facility to evaporation/percolation ponds and to adjacent lands for surface irrigation of fodder, fiber, and seed crops not intended for human consumption. Order NO. 72-178 is neither adequate nor consistent with current plans and policies of the Board. Reclamation is regulated by separate order.

The facility consists of two inlets, one for municipal and the other for industrial discharge. Currently, the treatment facility is not receiving industrial waste. The treatment plant is split into two treatment trains, each consisting of a grit chamber, a primary floatation sedimentation basin, an activated sludge unit, and a secondary clarifier. The effluent is then sent to final clarifier before being discharged to six evaporation/percolation ponds. Sludge is processed by means of two sludge digesters and eight sludge drying basins.

Each treatment train has a design capacity of 0.9 mgd for a total plant capacity of 1.8 mgd.

Area ground water of good quality (EC = 580 umhos/cm) is about 80 feet from ground surface with a gradient to the southeast. Soils consist of Traver and Tujunga sandy loam.

The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply. Surface water drainage is to Berenda Slough, a Tributary of the San Joaquin River.

The beneficial uses of the San Joaquin River are municipal, industrial, and agricultural supply; recreation; esthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydroelectric power generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

Average precipitation and evaporation rates are about 11 and 62 inches per year, respectively.

The action to update waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act in accordance with Title 14, California Code of Regulations, Section 15301.

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